Multisource Remote Sensing Data Classification With Graph Fusion Network

Xingqian Du, Xiangtao Zheng, *Member, IEEE*, Xiaoqiang Lu, *Senior Member, IEEE*, and Alexander A. Doudkin

***Abstract*— The land cover classification has been an important task in remote sensing. With the development of various sensors technologies, carrying out classification work with multisource remote sensing (MSRS) data has shown an advantage over using a single type of data. Hyperspectral images (HSIs) are able to represent the spectral properties of land cover, which is quite common for land cover understanding. Light detection and ranging (LiDAR) images contain altitude information of the ground, which is greatly helpful with urban scene analysis. Current HSI and LiDAR fusion methods perform feature extraction and feature fusion separately, which cannot well exploit the correlation of data sources. In order to make full use of the correlation of multisource data, an unsupervised feature extraction-fusion network for HSI and LiDAR, which utilizes feature fusion to guide the feature extraction procedure, is proposed in this article. More specifically, the network takes multisource data as input and directly output the unified fused feature. A multimodal graph is constructed for feature fusion, and graph-based loss functions including Laplacian loss and t-distributed stochastic neighbor embedding (t-SNE) loss are utilized to constrain the feature extraction network. Experimental results on several data sets demonstrate the proposed network can achieve more excellent classification performance than some state-of-the-art methods.**

***Index Terms*— Classification, deep learning, hyperspectral image (HSI), light detection and ranging (LiDAR), remote sensing.**

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Xingqian Du is with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China.

Xiangtao Zheng and Xiaoqiang Lu are with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China (e-mail: xiangtaoz@gmail.com).

Alexander A. Doudkin is with the Laboratory of System Identification, United Institute of Informatics Problems of the National Academy of Sciences of Belarus, 220012 Minsk, Belarus, and also with the Department of Computers, Belarusian State University of Informatics and Radioelectronics (BSUIR), 220012 Minsk, Belarus.

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I. Introduction

With the rapid development of sensor technologies, different kinds of remote sensing images have attracted much attention, such as hyperspectral image (HSI) and light detection and ranging (LiDAR). HSI contains hundreds of spectral channels through a wide range of electromagnetic spectrum which is greatly helpful for land cover analysis. Based on HSI, there are many remote sensing image processing and analysis applications, such as land cover pixel-wise classification [1]–[3], image restoration [4], and spectral unmixing [5]–[7]. LiDAR is able to acquire accurate elevation by illuminating laser light and accept the reflection [8], [9]. Due to the characteristics of multisource data, it is beneficial to identify land cover categories by combining HSI and LiDAR [9]–[11], which is already widely utilized in many applications such as land cover classification, forest monitoring [12], tree species mapping, and tree classification [13].

Recently, many multisource remote sensing (MSRS) data classification methods have been proposed to fuse HSI and LiDAR. To combine the multiple characteristics from HSI and LiDAR, feature extraction and feature fusion are necessary steps. Generally, multiple features are first extracted from HSI and LiDAR data, then land cover categories are identified by fusing the multiple features.

In order to exploit diverse information from multisource data and perform dimension reduction [14], a number of feature extractors have been proposed. Morphological filters [15] and convolutional neural network (CNN) are the most widely utilized feature extractors. Morphological profile (MP), attribute profile (AP), and extinction profile (EP) [16] are frequently used in the early works. For both HSI and LiDAR, morphological features are based on morphological openings and closings with a structural element of predefined size and shape. As for CNN, some CNN-based methods extract features with individual CNN branches for HSI and LiDAR [17], [18], others merge HSI and LiDAR into a single image before feeding into CNN [10], [19]. Chen *et al.* [17] utilized two CNN streams and concatenated the features from both streams for pixel-wise recognition. Xu *et al.* [18] developed a complicated network that includes spatial–spectral tunnels for HSI and cascade blocks for LiDAR. Although extracted features carry key information with less redundancy, feature fusion can be rough. Many methods fuse features by stacking them together [16]–[18]. In these methods, extracted features are bale to represent origin data in a lower dimension with

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less redundant information, but the correlation between data sources is ignored. For MSRS data classification, the correlation and complementary information between data sources should be well exploited.

To fuse the features extracted from multisource data effectively, some feature fusion strategies are proposed such as low-rank model and composite kernel, graph-based model [20], *etc.* Most of these methods first extract features, then fuse the features with various techniques. Low-rank model methods convert multisource features into a common space using low-rank sparse representation [21]–[24]. For example, Zhang and Prasad [24] proposed to fuse multisource features by imposing an adaptive source-specific weight on a joint sparse regularization term. Rasti *et al.* [21] proposed to fuse EPs with orthogonal total variation component analysis (OTVCA) in a lower dimension space. The composite kernel methods transfer multisource features into kernel space, such as Hilbert space [25], and fusion is conducted in the kernel space. Feature fusion strategies convert multisource features into a unified fused feature, but the feature extraction and fusion are separate and the original information contained in the features may be changed.

In order to resolve the problems mentioned above, a multisource graph fusion network is proposed for feature extraction and fusion. In the proposed network, feature extraction and feature fusion are integrated into a single feature extraction-fusion network. Meanwhile, the correlation in multisource data is utilized to guide the feature extraction process. First, a multimodal graph is constructed from HSI and LiDAR. Then, multimodal images are inputted into a feature extraction-fusion network. Finally, under the guidance of multimodal graph-based loss, a unified fused feature is outputted from the network for classification. Specifically, the multimodal graph is built by fusing the weight matrices of different data sources, which guarantees the complementary information of HSI and LiDAR is taken into account. Based on the multimodal graph, two graph embedding techniques including Laplacian embedding (LE) and t-distributed stochastic neighbor embedding (t-SNE) are utilized to perform feature learning and dimension reduction. LE ensures the pixels close to each other in the origin image are mapped close to each other in the feature space, preserving the local relationship and t-SNE preserves local similarities between pixels. The loss function of the proposed network is based on the two graph embedding techniques. The output of the network is a unified feature containing the correlation of multisource data.

In summary, the contributions of this work can be listed as follows:

1) A graph fusion network is proposed to extract a fused feature from multisource data. Both feature extraction and feature fusion are integrated into a single feature extraction-fusion network.
2) To capture the correlation between HSI and LiDAR, a multimodal graph is built by fusing the weight matrices of different data sources. The multimodal graph is utilized to guide the feature extraction process for MSRS classification.

The remaining parts of this article are organized as follows. Section II discusses the related work. Section III describes the proposed network in detail. The experiments are shown in Section IV.

II. Related Work

A. HSI and LiDAR Classification

Similar to HSI classification [26], HSI and LiDAR classification task employs coregistered images of two data sources for pixel-wise land cover recognition. Some HSI and LiDAR classification methods focus on feature extraction and some develop various feature fusion strategies.

MPs and MP-based features, such as APs and EPs [27], are widely used in early works [28], [29]. For example, Benediktsson *et al.* [28] proposed extended MP for pixel-wise classification. Debes *et al.* [22] applied morphological openings and closings with partial reconstruction on LiDAR and first two principle components (PCs) of HSI. Liao *et al.* [30] extracted MPs then classified with Support Vector Machine (SVM), and decision fusion is performed for pixel-wise recognition. Khodadadzadeh *et al.* [31] concatenated extended APs and carried out classification with subspace multinomial logistic regression classifier. Furthermore, Ge *et al.* [32] extracted extended multi-EP [21], [27] and local binary pattern (LBP) from multisource data, then performed feature fusion with residual fusion strategy.

Due to the powerful capability of extracting deep features, CNN-based methods have been exploited in the field of LiDAR and HSI fusion in recent years. Morchhale *et al.* [33] fused HSI and LiDAR at pixel-level and utilized CNN for classification. In order to make full use of the deep feature of multisource data, Chen *et al.* [17] designed two separate CNN tunnels to extract features of HSI and LiDAR, respectively. The features extracted from CNN tunnels are stacked and fed into fully connected (FC) layers and logistic regression classifier. Xu *et al.* [18] designed a more complex CNN architecture especially for multisource classification, named two-branch CNN (TBCNN), which consists of an HSI branch and a LiDAR branch. The former extracts spatial and spectral information from HSI, and the latter utilizes specially designed cascade blocks to extract the feature of LiDAR. Furthermore, Zhao *et al.* [10] designed a pixelwise affinity branch for LiDAR feature extraction and a hierarchical random walk layer for enforcing spatial consistency in the deeper layers of the network.

B. Graph Embedding

For HSI, each pixel can be represented as a vertex in an undirected graph and the similarity among pixels can be represented as edge connections. Thus, graph embedding techniques can be applied to HSI classification [34] and attracted much attention. Gong *et al.* [35] utilized graph embedding to generate loss function for HSI feature extraction. Cahill *et al.* [36] proposed a dimension reduction and clustering approach named Spatial–Spectral Schroedinger Eigenmaps (SSSE) for HSI classification. Qin *et al.* [37] proposed spectral–spatial graph convolutional networks (S^2GCN) for HSI on the basis

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of graph convolutional networks (GCN) [38]. Furthermore, Hong *et al.* [39] proposed a minibatch graph convolutional network (miniGCN) to reduce the computational cost when using GCN for HSI classification. Shukla *el al.* [40] proposed a dimension reduction technique Laplacian embedding based t-distributed stochastic neighbor embedding (LEt-SNE) by combining LE and t-SNE on the graph constructed from HSI. Graph embedding can also be used to refine HSI features. For example, Liu *et al.* [41] used graph embedding to refine feature extracted by an autoencoder and the embeddings are fed to a classifier.

Graph embedding is also used in the fusion of HSI and LiDAR. The most straightforward and common way is building graphs for each feature and conduct element-wise multiplication among all the adjacency matrices to generate a new adjacency matrix [22], [27], [30]. Liao *et al.* [29] proposed semisupervised graph-based fusion (SSGF) which takes both labeled and unlabeled samples into account. Multimodal edge weights is proposed [42] for building a graph containing information of multisource images. Gu and Wang [43] proposed a discriminative graph-based fusion strategy, which generates a similarity matrix and a dissimilarity matrix for each feature.

C. Remote Sensing Image Fusion

Image fusion task in remote sensing combines the advantages of different data sources for a better understanding of the Earth's surface. For a specific land cover analysis task, the fusion technique can be performed at three different processing levels according to the stage at which the image fusion takes place: pixel level, feature level, and decision level [44].

Pixel level image fusion generates a single image from geometrically registered images of the same scene. A typical instance of pixel-level fusion is pansharpening, which is the fusion of the panchromatic image (PANI) and multispectral image (MSI) [45]. Similar to pansharpening for PANI and MSI [46], Chen *et al.* [19] utilized PC Spectral Sharpening to fuse HSI and LiDAR before feeding the image into CNN.

Feature level fusion methods perform feature extraction on each data source independently, then fuse features into a unified feature with various techniques for further analysis. Features to be fused may include various types such as spectral, spectral, and textural [47]. Golipour *et al.* [48] proposed to integrate hierarchical segmentation results into Markov random field spatial prior in the Bayesian framework for spectral–spatial HSI classification. Hong *et al.* [49] proposed a generalized MSRS classification framework by introducing five plug-and-play fusion modules based on different feature fusion strategies.

Decision level fusion merges results from several individual data sources after each data source has undergone a preliminary classification. Specifically, in multisource land cover classification tasks, decision level methods carryout the whole classification procedure with each kind of data source, then fuse the output class maps into the final result using certain strategies, such as voting [30] and Bayesian inference.

III. Proposed Method

This section details the proposed unsupervised feature extraction-fusion network. Section III-A introduces the network architecture, Section III-B introduces the construction of the multimodal graph, and Section III-C details the employed graph embedding techniques and loss function.

A. Network Architecture

The framework of the proposed method is shown in Fig. 1. The input is multisource data and the output is a unified fused feature. Specifically, HSI and LiDAR are fed into a feature extraction-fusion network, then feature fusion and extraction are carried out in the single network. To take advantage of the complementary information and correlation between HSI and LiDAR, an adjacency matrix of a multimodal graph is computed by fusing the weight matrices of HSI and LiDAR. Moreover, a graph-based loss function is introduced to constrain the feature extraction-fusion network from the perspective of graph embedding. When the feature is extracted, an SVM classifier is employed to carry out a classification task.

Each pixel in the data set is represented as a vector and the network takes the vectors of the whole image as input. For a coregistered HSI and LiDAR data set, let $\mathbf{X^H}$ and $\mathbf{X^L}$ denote HSI and LiDAR image, respectively. Specifically

$$\mathbf{X^H} = \left\{\mathbf{x_1^H}, \mathbf{x_2^H}, \ldots, \mathbf{x_n^H}\right\}, \quad \mathbf{x_i^H} \in \mathbb{R}^{\mathbf{C}} \tag{1}$$

where $\mathbf{x_i^H}$ is the vector representing the ith pixel, n is the number of pixels in the image and C is the number of spectral channels of HSI. Similarly

$$\mathbf{X^L} = \left\{\mathbf{x_1^L}, \mathbf{x_2^L}, \ldots, \mathbf{x_n^L}\right\}, \quad \mathbf{x_i^L} \in \mathbb{R}^{\mathbf{1}}. \tag{2}$$

Note that LiDAR image only has one channel. Thus, the input of the network is

$$\mathbf{X} = \{\mathbf{x_1}, \mathbf{x_2}, \ldots, \mathbf{x_n}\}, \quad \mathbf{x_i} \in \mathbb{R}^{\mathbf{C+1}} \tag{3}$$

where $\mathbf{x_i} = \mathrm{CAT}(\mathbf{x_i^H}, \mathbf{x_i^L})$ and $\mathrm{CAT}(\cdot, \cdot)$ represents concatenating operation. Then, $\mathbf{X}$ is fed into the feature extraction-fusion network.

In order to show the potential ability of the multimodal graph and graph-based loss, FC layers are employed as the backbone of the proposed network. The reason is that the multimodal graph and graph-based loss play a major role in feature extraction and fusion, and the network structure is not the emphasis of the research. Thus, we choose a shallow and robust structure as the feature extraction network. Specifically, the network consists of two FC layers and one batch normalization (BN) layer. Note that the BN layer is after FC layers, there is no BN operation for the multimodal graph. In this article, a rectified linear unit (ReLU) is employed as an activation function. Since the input of the network are multimodal images, in practice, the backbone of the feature-extraction network can be replaced with other networks, such as convolutional layers. After training with the loss function based on a multimodal graph, a unified fused feature is outputted and sent to the classifier for pixel-wise recognition.

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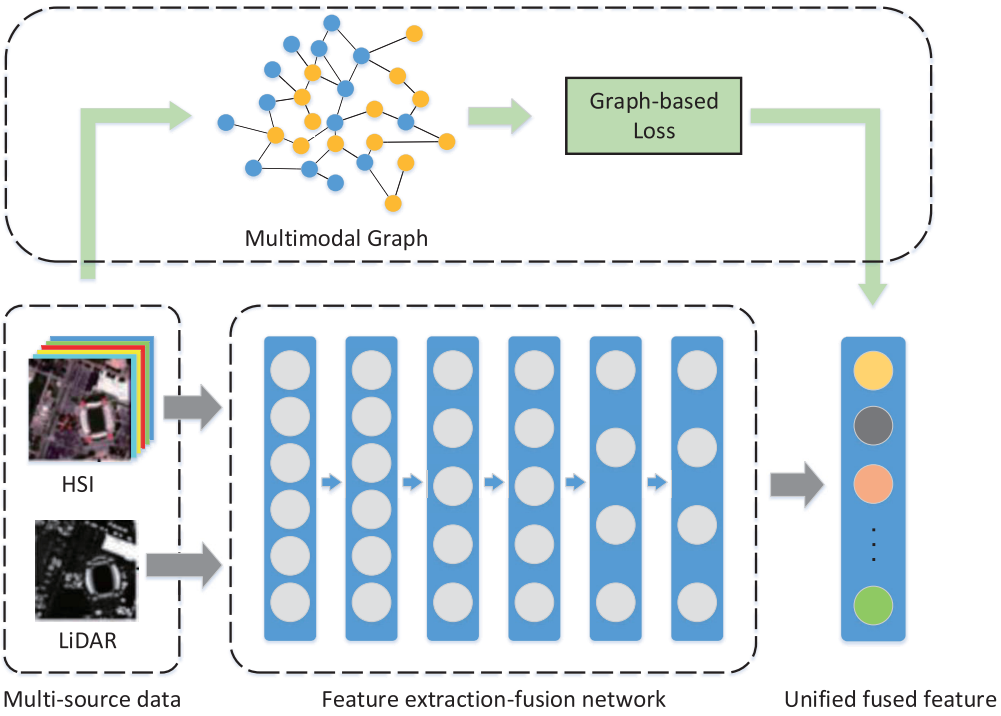


Fig. 1. Overall framework of the proposed method. As shown in the figure, multisource images are inputted into the feature extraction-fusion network, and a unified fused feature is outputted. Meanwhile, a multimodal graph is built from multisource images. After building the multimodal graph, graph embedding techniques are performed to generate graph-based loss function, which guides the feature extraction-fusion process.

B. Multimodal Graph

In order to effectively fuse the information between multimodal images, the graph structure is employed to represent the input pixels. Compared to CNN, graph structure has a stronger ability to represent a relationship among all vertices since the size of the convolution kernel limits CNN's global information extraction capability. An image can be converted into a graph by considering each pixel as a vertex, and the edges between the vertices can be calculated by the similarity between pixels. When we have images in two modals, a multimodal graph could be built by fusing graphs of each modal. The rest of this section will specify the building of the multimodal graph.

Based on the multisource data $\mathbf{X^H}$ and $\mathbf{X^L}$, an undirected graph $G = (V, E)$ can be constructed, where V and E represent the vertices and edges, respectively. Each vertex in the graph corresponds to a pixel of the origin data set. The adjacency matrix, which represents the edge connections of the graph, depends on the weight matrix in HSI $\mathbf{W}^H = \{w_{ij}^H\} \in \mathbb{R}^{n\times n}$ and that in LiDAR $\mathbf{W^L} = \{w_{ij}^L\} \in \mathbb{R}^{n\times n}$, where n denotes the number of pixels. For HSI, we define the weights of the edges as follows:

$$w_{ij}^H = \mathrm{DIST}\left(\mathbf{x_i^H}, \mathbf{x_j^H}\right) \tag{4}$$

where w_{ij}^H represents the weight between the ith pixel and the jth pixel. The function $\mathrm{DIST}(\cdot,\cdot)$ denotes the distance between $\mathbf{x_i^H}$ and $\mathbf{x_j^H}$. The distance metric of $\mathrm{DIST}(\cdot,\cdot)$ in this article is employed as Euclidean distance. Similarly, for LiDAR, we have

$$w_{ij}^L = \mathrm{DIST}\left(\mathbf{x_i^L}, \mathbf{x_j^L}\right). \tag{5}$$

In order to fuse the information in HSI and LiDAR, we define the fused multimodal adjacency matrix by

$$\mathbf{A} = \mathrm{KNN}(\alpha \mathbf{W}^H + \beta \mathbf{W}^L) \tag{6}$$

where α and β are parameters for HSI and LiDAR weight matrices. Meanwhile, the k-nearest neighborhood (KNN) strategy is employed. The motivation behind building the multimodal graph is enlarging the difference between categories in both HSI and LiDAR. More specifically, for example, consider pixel $\mathbf{x_i}$ and $\mathbf{x_j}$ with similar material (such as grass and trees), they have a small distance in the spectral domain of HSI but a large distance in LiDAR (significant elevation difference). By introducing $\mathbf{W}^L$ to the adjacency matrix, the difference between these two classes are enlarged and they are more likely to be correctly classified compared to using $\mathbf{W}^H$ alone.

C. Loss Function

When the multimodal graph containing correlation of multisource data is built in Section III-B, graph embedding techniques are employed for generating loss function. To preserve the local neighborhood information in the multimodal graph, LE is employed. Considering that LE has limitations when applied to HSI, such as lack of consideration of natural spatial correlation among pixels [50], graph embedding technique combining t-SNE and LE [40] is adopted.

The overall loss function for the feature extraction-fusion network is in the form of 7, which includes two parts

$$\mathrm{Loss} = kL_{\mathrm{Laplacian}} + L_{t-\mathrm{SNE}}. \tag{7}$$

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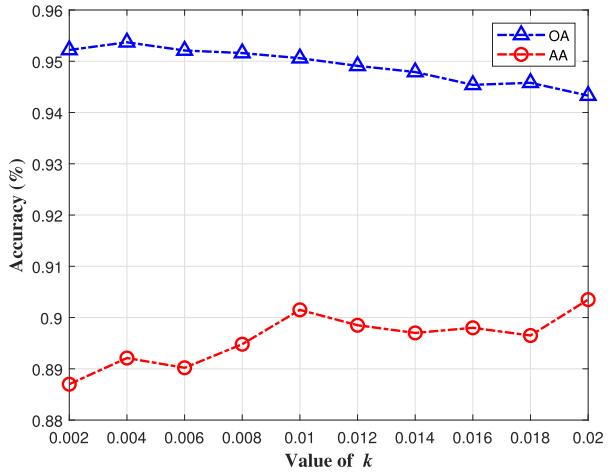


Fig. 2. Classification performance with different values of k on MUUFL data set.

The first term is traditional Laplacian loss, and the latter is t-SNE loss. The loss function will be explained in detail as follows.

1) Laplacian Embedding: To preserve the natural clusters in the multimodal graph, LE is introduced as the first loss function for feature extraction.

Based on the multimodal graph built in Section III-B, the Laplacian matrix can be calculated by

$$\mathbf{L} = \mathbf{D} - \mathbf{A} \tag{8}$$

where $\mathbf{D}$ is the diagonal degree matrix defined by $\mathbf{D}_{ii} = \sum_j \mathbf{A}_{ij}$. Considering the large entries in $\mathbf{A}$ represent lower distance of vertices and small entries represent farther distance, we minimize $L_{\text{Laplacian}}$ to further constrain the vertices with lower distance

$$L_{\text{Laplacian}} = \sum_{i,j} (\mathbf{y_i} - \mathbf{y_j})^2 \mathbf{A}_{ij} = 2\mathbf{Y}^T \mathbf{L}\mathbf{Y}. \tag{9}$$

$\mathbf{Y}$ represents the output of the feature extraction network.

2) t-SNE: To exploit the information contained in HSI and LiDAR, t-SNE combined with LE is introduced due to its ability to preserve local similarities between pixels and stretching intersample distances [40].

Let $\mathbf{X} = \{\mathbf{x_i}\} \in \mathbb{R}^n$ be the input pixels of the data set, then we have the probability of $\mathbf{x_j}$ being a neighbor of $\mathbf{x_i}$ is defined by t-SNE in 10

$$p_{j|i} = \frac{\exp\left(-\|\mathbf{x_i} - \mathbf{x_j}\|^2 / 2\sigma_i^2\right)}{\sum_{k \neq i} \exp\left(-\|\mathbf{x_i} - \mathbf{x_k}\|^2 / 2\sigma_i^2\right)}. \tag{10}$$

The variance σ_i is a loose interpretation of the density of samples around $\mathbf{x_i}$, smaller values of σ_i are used in denser parts of the data space. In order to stretch intersample distances to perform dimensionality reduction, a compression factor (CF) uses the adjacency matrix A is introduced [40], which can make full use of the distance between samples in X. As in 11, $\tilde{p}_{j|i}$ takes account of the connections among the vertices in the multimodal graph and takes the place of $p_{j|i}$ in the optimization of t-SNE

$$\tilde{p}_{j|i} = \frac{p_{j|i}\{(\text{CF} - 1)A_{ij} + 1\}}{\sum_j p_{j|i}\{(\text{CF} - 1)A_{ij} + 1\}}. \tag{11}$$

TABLE I
TRAINING AND TESTING SAMPLES OF TRENTO DATA SET

No.	Class	Train	Test	All
1	Apple trees	404	3630	4034
2	Buildings	291	2612	2903
3	Ground	48	431	479
4	Woods	913	8210	9123
5	Vineyard	1051	9450	10501
6	Roads	318	2856	3174
	Total	3025	27189	30214

Similar to 10, a t-distribution $q_{j|i}$ is computed for the lower dimension y, which is the output of feature extraction

$$q_{ij} = \frac{\exp(-\|\mathbf{y_i} - \mathbf{y_j}\|^2)}{\sum_{k \neq i} \exp(-\|\mathbf{y_i} - \mathbf{y_k}\|^2)}. \tag{12}$$

By minimizing the Kullback–Leibler (KL) divergence between $\tilde{p}_{j|i}$ and $q_{j|i}$, we have the t-SNE loss function in the form of

$$L_{\text{t-SNE}} = \sum_i \sum_j \tilde{p}_{i|j} \log \frac{\tilde{p}_{i|j}}{q_{i|j}}. \tag{13}$$

IV. EXPERIMENTS

To validate the effectiveness of the proposed network, experiments are conducted on three data sets. Section IV-A introduces the data sets used in the experiments. Section IV-B lists all the compared methods in this article. The experiment settings are detailed in Section IV-C, and the complete experiment results are shown in Section IV-D.

A. Data Sets

Considering the diversity of data sets, we choose three coregistered HSI and LiDAR data sets that vary in the volume of data: Trento, Missouri University and University of Florida (MUUFL) Gulfport, and 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest. The detailed information of the data sets is introduced as follows.

1) Trento Data Set: The data were captured over the rural area of the city of Trento, Italy, with a resolution of 600 × 166. The HSI in the Trento data set consists of 63 bands ranging from 420.89 to 989.09 nm where spectral resolution is 9.2 nm, and the spatial resolution is 1 m. There are six classes including apple trees, buildings, ground, woods, vineyard, and roads. The number of samples in each class is shown in Table I.

2) MUUFL Gulfport Data Set: The MUUFL Gulfport data set was collected in November 2010 over the University of Southern Mississippi Gulf Park Campus, located in Long Beach, Mississippi. The data collection contains coregistered hyperspectral and LiDAR data over the campus. The original MUUFL Gulfport data set contains 325 × 220 pixels across 64 bands. There are 11 classes including trees, buildings, water, *etc.* Table II shows the detailed class information and training testing samples of the MUUFL data set.

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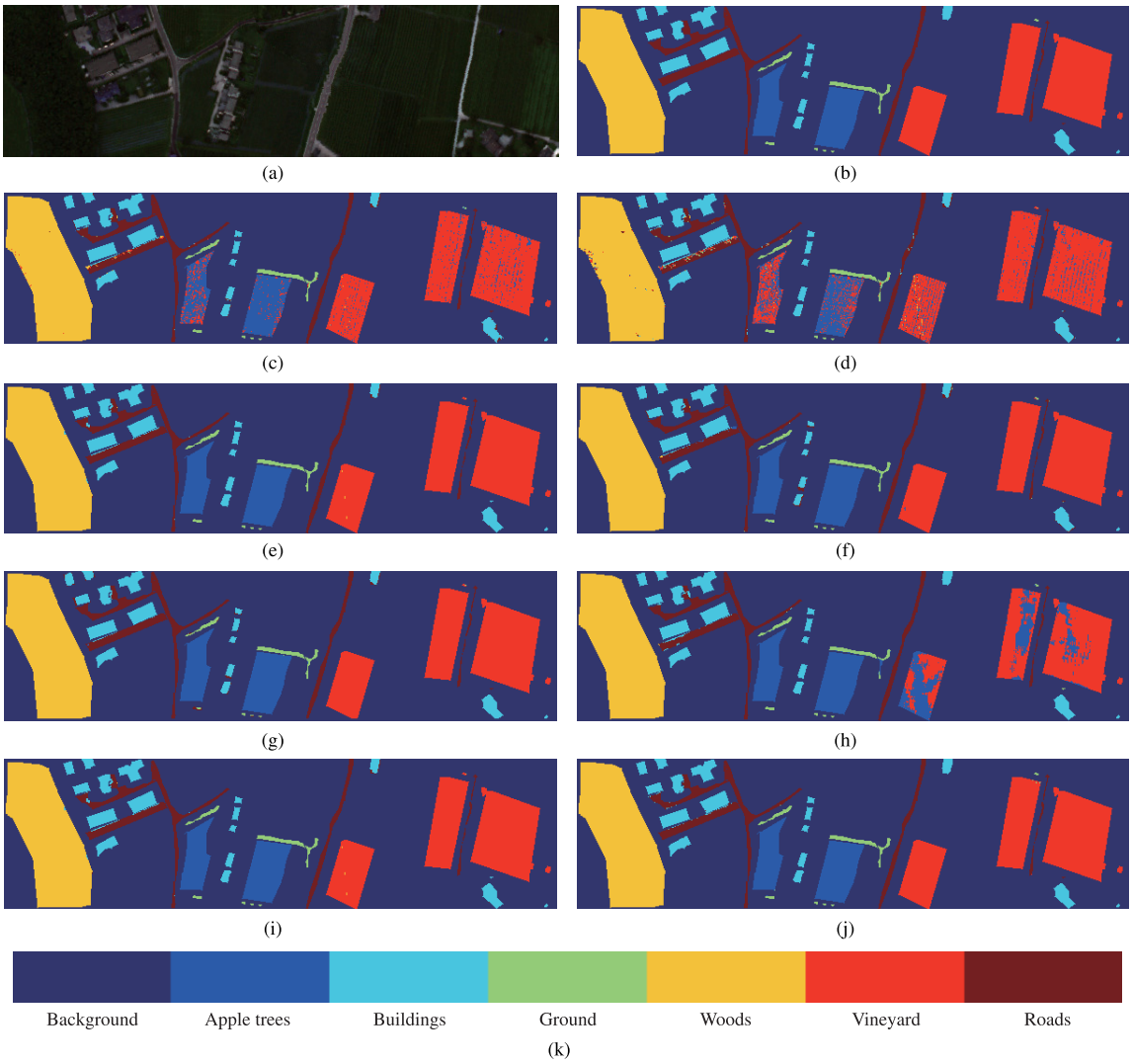


Fig. 3. Classification results on Trento data set. (a) Pseudo color image with the 4th, 18th and 30th band of HSI. (b) Ground truth. (c) SVM. (d) LE. (e) SSSE. (f) CCNN. (g) 3-D CNN. (h) TBCNN. (i) HRWN. (j) The proposed method. (k) Legend of different classes.

TABLE II
TRAINING AND TESTING SAMPLES OF MUUFL DATA SET

No.	Class	Train	Test	All
1	Trees	2325	20921	23246
2	Grass ground	427	3843	4270
3	Mixed ground	689	6193	6882
4	Dirt and sand	183	1643	1826
5	Road	669	6018	6687
6	Water	47	419	466
7	Buildings	224	2009	2233
8	Shadow	624	5616	6240
9	Sidewalk	139	1246	1385
10	Yellow curb	19	164	183
11	Cloth panels	27	242	269
	Total	5373	48314	53687

3) 2013 IEEE GRSS Data Fusion Constest Data Set: The scene of the 2013 Data Fusion Contest (DFC2013) data set was acquired in June 2012 by an airborne sensor over the area of University of Houston campus and neighbor area. The HSI and a LiDAR image are both at the same spatial resolution (2.5 m). The hyperspectral imagery consists of 144 spectral bands in the 380 to 1050 nm region. The corresponding coregistered LiDAR image consists of elevation in meters above sea level. The data were acquired by the NSF-funded Center for Airborne Laser Mapping (NCALM) over the University of Houston campus and the neighboring urban area. There are 15 classes in the data set. The number of samples in each class is shown in Table III.

B. Compared Methods

In order to assess the performance of the proposed network, comparative experiments are conducted on following HSI and LiDAR classification methods:

1) *Origin Data With SVM Classifier:* Classifying merged HSI and LiDAR without feature extraction. As the baseline of other methods, an SVM classifier is directly trained with the spectral information of HSI and single-band LiDAR.

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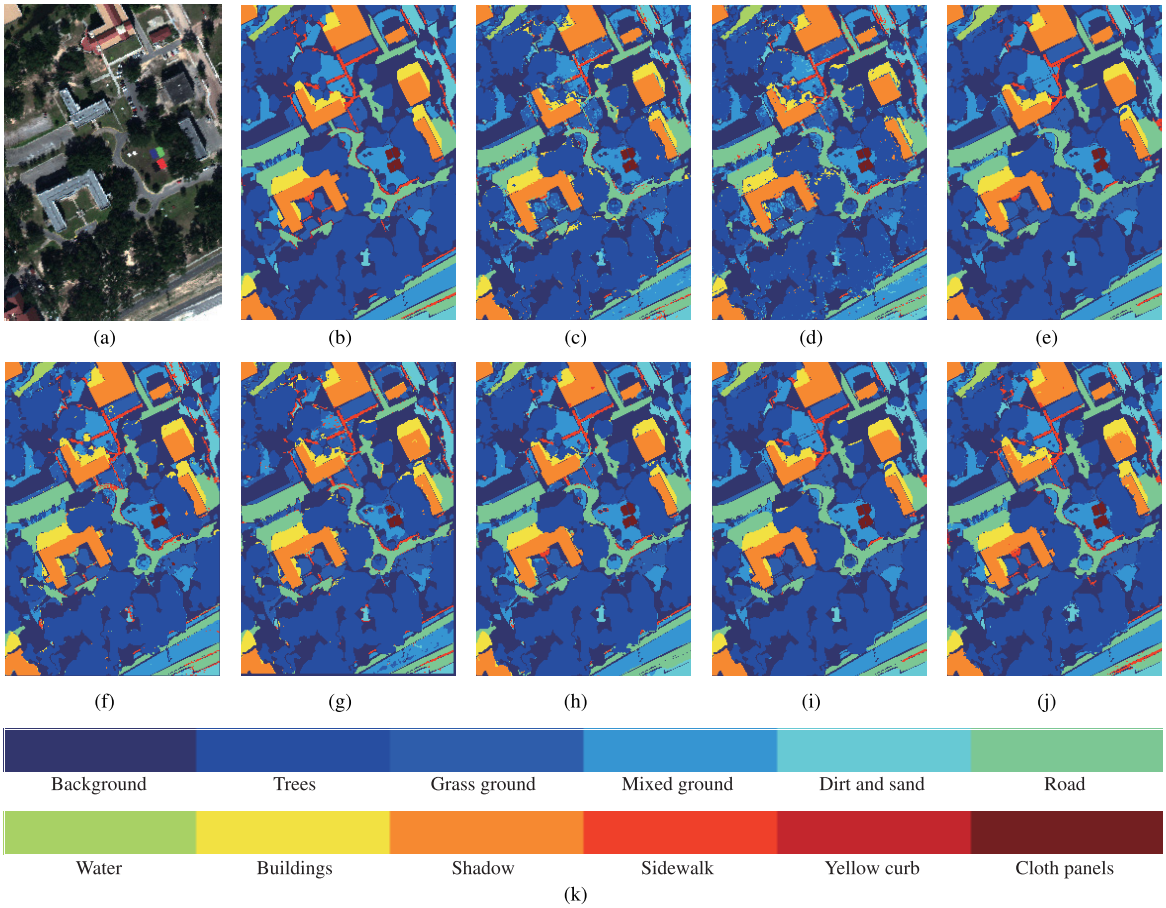


Fig. 4. Classification results on MUUFL data set. (a) Color image. (b) Ground truth. (c) SVM. (d) LE. (e) SSSE. (f) CCNN. (g) 3-D CNN. (h) TBCNN. (i) HRWN. (j) Proposed method. (k) Legend of different classes.

TABLE III
TRAINING AND TESTING SAMPLES OF DFC2013 DATA SET

No.	Class	Train	Test	All
1	Healthy grass	126	1125	1251
2	Stressed grass	126	1128	1254
3	Synthetic grass	70	627	697
4	Trees	125	1119	1244
5	Soil	125	1117	1242
6	Water	33	292	325
7	Residential	127	1141	1268
8	Commercial	125	1119	1244
9	Road	126	1126	1252
10	Highway	123	1104	1227
11	Railway	124	1111	1235
12	Parking lot 1	124	1109	1233
13	Parking lot 2	47	422	469
14	Tennis court	43	385	428
15	Running track	66	594	660
	Total	1510	13519	15029

2) *LE:* A similarity graph is constructed and LE is performed on the graph, then SVM is used for classification.
3) *SSSE [36]:* This method is a graph-based spatial-spectral dimensionality reduction method for multiband image classification. Schroedinger Eigenmaps on graphs defined with spectral information is combined with LE. SVM is employed as the classifier.
4) *Contextual CNN (CCNN) [51]:* The backbone of CCNN is a 9-layer deep fully convolution network (FCN). The network contains a multiscale filter bank exploiting spectral and contextual information and residual connections which makes the network deeper.
5) *3-D CNN [52]:* The 3-D CNN takes a 3-D patch as an input and simultaneously process the spatial and spectral components with four 3-D convolution layers.
6) *TBCNN [18]:* A representative CNN architecture for HSI and LiDAR classification. Two CNN branches with different architectures extract features from different data sources, and the features are fused by concatenation for classification.
7) *Hierarchical Random Walk Network (HRWN) [10]:* Two CNN branches are used to extract spatial-spectral feature of HSI and LiDAR. A hierarchical random walk layer based on the LiDAR weight graph is introduced as the classifier.

C. Experimental Settings

The experiments are implemented with the tensor flow in the environment of python 3. In order to show the validity of the proposed network, we employ only simple FC layers as the

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON TRENTO DATA SET

Method	SVM	LE-SVM	SSSE	CCNN	3D CNN	TBCNN	HRWN	Proposed(H)	Proposed(H+L)
1	81.90%	60.22%	**100.00%**	**100.00%**	99.97%	53.83%	99.19%	99.81%	**100.00%**
2	96.94%	98.12%	99.08%	94.80%	96.53%	98.53%	93.79%	92.38%	**99.66%**
3	96.29%	93.04%	97.68%	96.06%	90.49%	**98.73%**	97.00%	95.13%	97.91%
4	99.67%	99.00%	99.84%	**100.00%**	**100.00%**	**100.00%**	99.91%	99.94%	**100.00%**
5	94.70%	89.96%	99.92%	99.95%	**100.00%**	99.95%	98.68%	99.81%	99.93%
6	94.78%	90.76%	97.06%	97.13%	98.73%	**99.86%**	98.70%	85.33%	97.62%
OA	94.74%	89.64%	99.49%	99.12%	99.38%	88.36%	98.60%	97.54%	**99.68%**
AA	94.05%	88.52%	98.93%	97.99%	97.62%	91.82%	97.88%	95.40%	**99.24%**

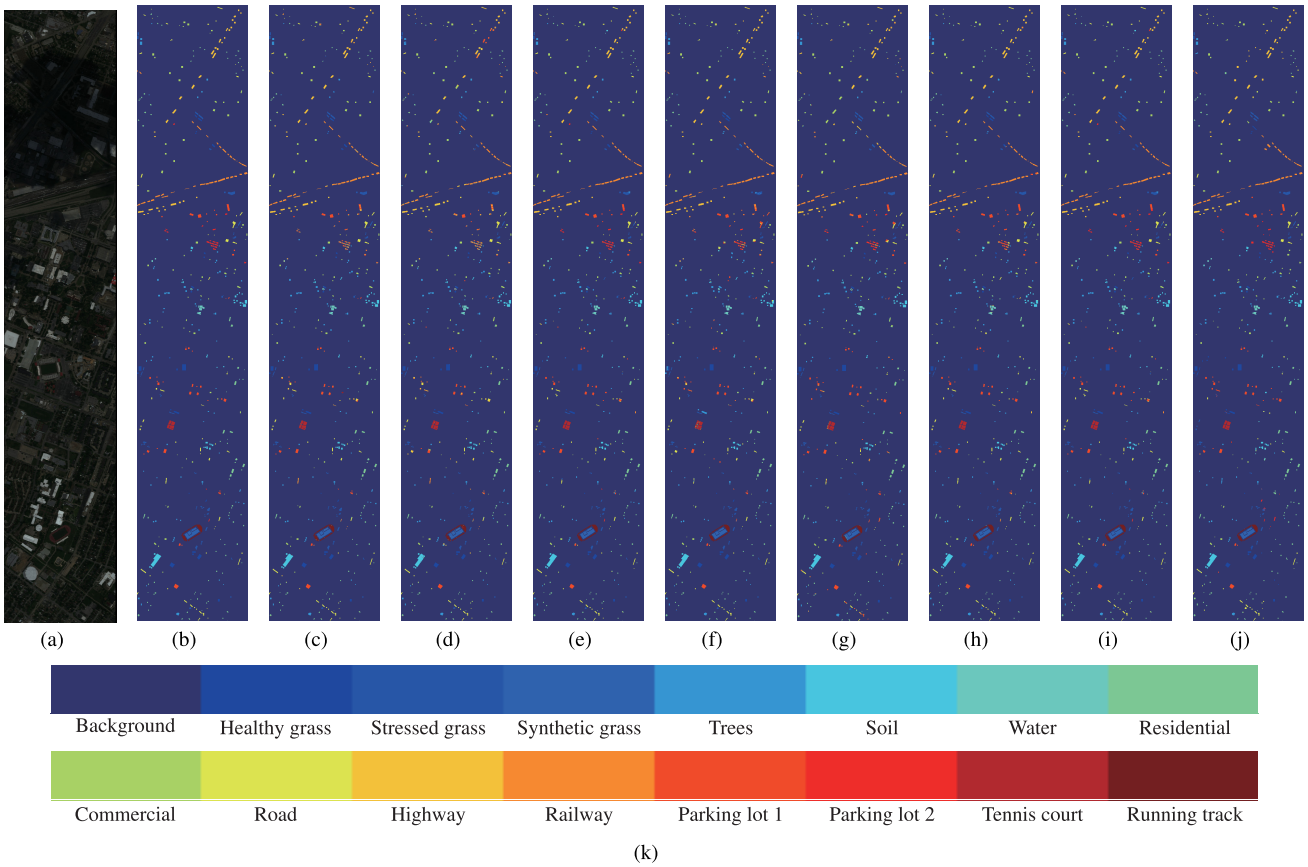


Fig. 5. Classification results on DFC2013 data set. (a) Pseudo color image with the 59th, 40th, and 23rd band of HSI. (b) Ground truth. (c) SVM. (d) LE. (e) SSSE. (f) CCNN. (g) 3-D CNN. (h) TBCNN. (i) HRWN. (j) Proposed method. (k) Legend of different classes.

backbone of the feature extraction network. There are two FC layers, and the first layer has 64 hidden units, the second has 50. While building the fused graph, the neighbor size of KNN is employed with 10. The weights of HSI, LiDAR, and spatial information for fusing bands are 1, 0.7, and 0.2, respectively. For the loss function, k is set to 0.01. The batch size is set to 1024, and the model is trained for 300 epochs with the learning rate at 0.001. For SVM, the dimension of feature is the sum of the spectral dimension of HSI and LiDAR. In the proposed method, LE and SSSE, the dimensions of the extracted feature are all set to 50. For deep learning-based methods and correlation-based methods (CCNN, 3-D CNN, TBCNN, and HRWN), dimensions of features are consistent with the origin settings. For a fair comparison, data augmentation operation (such as flipping and rotating of samples) is disabled in all methods. We adopt overall accuracy (OA) and average accuracy (AA) as the evaluation indicators of the experiment results. OA represents the ratio of the correctly predicted samples and all samples while AA is the average accuracy of all classes.

D. Experiment Results

Table IV and Figure 3 show the classification results on Trento data set. As is shown in Table IV, the proposed method

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TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON MUUFL DATA SET

Method	SVM	LE	SSSE	CCNN	3D CNN	TBCNN	HRWN	Proposed(H)	Proposed(H+L)
1	95.21%	92.58%	**98.03%**	97.81%	96.54%	97.16%	96.79%	95.95%	96.20%
2	64.87%	71.95%	93.08%	83.35%	**99.15%**	85.71%	69.81%	96.62%	97.46%
3	83.87%	78.14%	93.25%	90.02%	46.90%	81.15%	**95.14%**	93.11%	93.71%
4	71.09%	74.13%	91.66%	81.86%	85.80%	92.25%	93.21%	92.57%	**93.90%**
5	91.86%	90.13%	**96.91%**	94.84%	94.42%	90.00%	93.79%	94.25%	95.47%
6	91.41%	73.51%	95.23%	87.83%	89.31%	**98.33%**	95.98%	98.57%	96.90%
7	43.40%	67.99%	91.94%	91.39%	92.19%	94.57%	84.80%	92.04%	**95.38%**
8	92.82%	79.31%	96.97%	96.96%	94.48%	97.25%	**98.57%**	93.86%	95.06%
9	18.38%	45.02%	58.67%	72.02%	76.71%	85.54%	**84.86%**	79.53%	79.41%
10	78.05%	3.66%	1.22%	32.73%	25.63%	56.67%	**85.37%**	48.17%	54.82%
11	66.53%	87.19%	**94.21%**	83.88%	62.40%	95.93%	88.65%	90.91%	93.39%
OA	85.46%	83.87%	94.90%	93.35%	88.48%	92.56%	92.60%	94.32%	**95.06%**
AA	72.50%	69.42%	82.83%	82.97%	78.50%	88.60%	89.73%	88.69%	**90.15%**

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON DFC2013 DATA SET

Method	SVM	LE-SVM	SSSE	CCNN	3D CNN	TBCNN	HRWN	Proposed(H)	Proposed(H+L)
1	93.33%	90.40%	95.47%	73.45%	86.23%	88.85%	91.09%	96.80%	**99.47%**
2	98.05%	95.57%	93.97%	94.77%	99.91%	96.50%	95.65%	95.74%	**100.00%**
3	98.72%	98.41%	99.52%	99.52%	97.45%	**100.00%**	99.86%	100.00%	**100.00%**
4	99.20%	94.92%	96.69%	99.91%	98.13%	99.55%	**100.00%**	96.34%	95.89%
5	99.55%	95.40%	**100.00%**	94.90%	**100.00%**	99.64%	98.03%	99.02%	**100.00%**
6	90.75%	85.27%	97.95%	88.01%	88.36%	99.59%	**100.00%**	99.66%	**100.00%**
7	91.59%	83.51%	97.46%	97.63%	79.49%	97.46%	96.74%	95.79%	**98.25%**
8	91.96%	89.31%	94.10%	86.52%	96.25%	**100.00%**	98.29%	91.06%	94.82%
9	84.55%	69.16%	93.43%	64.51%	56.08%	88.13%	97.11%	94.49%	96.45%
10	94.02%	53.24%	98.19%	99.37%	75.54%	89.72%	84.00%	99.82%	**100.00%**
11	90.82%	77.77%	**100.00%**	98.65%	83.72%	91.21%	93.33%	99.28%	**100.00%**
12	85.39%	54.40%	89.81%	73.51%	93.15%	76.66%	79.89%	93.24%	**97.29%**
13	23.46%	15.60%	94.79%	44.08%	50.95%	**100.00%**	92.76%	61.85%	80.57%
14	97.40%	90.95%	**100.00%**	92.73%	96.62%	**100.00%**	97.42%	100.00%	**100.00%**
15	97.81%	92.46%	**100.00%**	99.83%	99.49%	**100.00%**	95.51%	99.83%	**100.00%**
OA	91.26%	79.53%	96.38%	88.08%	87.08%	93.29%	93.50%	95.61%	**97.91%**
AA	89.11%	79.09%	96.76%	87.16%	86.76%	95.15%	94.65%	94.86%	**97.52%**

TABLE VII
ABLATION EXPERIMENT OF THE PROPOSED METHOD ON MUUFL DATA SET

	Laplacian loss	t-SNE loss	OA	AA
HSI graph	✓	✓	94.34%	89.04%
LiDAR garph	✓	✓	87.26%	86.92%
Multi-modal graph	✓		62.34%	66.35%
Multi-modal graph		✓	94.78%	90.73%
Multi-modal graph	✓	✓	**95.06%**	**90.15%**

achieves the best performance in terms of OA and AA. OA of the proposed method is 99.68%. The proposed method has also reached the highest accuracy in three classes including apple trees, buildings, and vineyard. Compared to the baseline SVM, the proposed method has improved OA for 4.94%. TBCNN performs better on the third and the sixth class, which is ground and road. Compared to the results using HSI alone (H), the fusion of HSI and LiDAR (H + L) improves OA for 2.14%.

The classification results of the proposed method and other methods on the MUUFL data set are shown in Table V and Figure 4. It can be found in Table V that the proposed

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method performs better than all the compared methods. More specifically, the proposed method has the top OA (95.06%) and reached the highest accuracy in two classes, which are dirt and buildings. The 10th class (yellow curb) is more difficult to classify among all the eleven classes since there are too few samples for training (only 19). In the proposed method, most misclassified samples are recognized as mixed ground and road.

The results of DFC2013 data set are shown in Table VI and Figure 5. Compared to other methods, the proposed method reaches the highest accuracy in 11 of the 15 classes and the OA is 97.91%, which is 6.65% higher than the baseline SVM. Considering the accuracy of the 13th class in the results of the proposed method is lower than other state-of-the-art methods, we find that some of the samples in the 13th class (parking lot 2) are misclassified as the 12th class (parking lot 1). Compared with classifying with HSI alone, the fusion of HSI and LiDAR improves OA for 2.3% and AA for 2.66%.

To validate the function of the multimodal graph and graph-based loss in the proposed method, ablation experiments are conducted on the MUUFL data set. As shown in Table VII, the model achieves the highest accuracy in terms of OA when multimodal graph is adopted (the last row of Table VII) compared to building graph with a single data source (HSI graph and LiDAR graph in Table VII). The OA of the model when using t-SNE loss alone is 32.72% higher than using Laplacian loss alone.

The tradeoff between the LE loss and t-SNE loss is investigated by adjusting the parameter k. Classification results on MUUFL data set when k takes different values are shown in Fig. 2. As we can see in the results, when k drops below 0.008, AA decreases obviously while OA increases, which indicates that the Laplacian loss is helpful for improving the accuracy of a specific class.

V. Conclusion

In this article, a feature extraction–fusion network for multisource classification is proposed. In the proposed network, the correlation between multisource data is utilized to guide the process of feature extraction. Specifically, a multimodal graph is combined with feature extraction network to integrate feature extraction and fusion into a single extraction-fusion network and guide the feature extraction process with the inner correlation of multisource data. Moreover, LE and t-SNE are employed on the multimodal graph to perform feature learning and dimension reduction. Loss function of the network is based on the graph embedding techniques to constrain the feature extraction–fusion network. Experiments on several common remote sensing multisource data sets show the validity of the proposed network, and the comparison with classical methods and state-of-the-art methods shows that the proposed network has reached impressive accuracy.

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Xingqian Du is pursuing the Ph.D. degree with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, and also with the University of Chinese Academy of Sciences, Beijing, China.

His research interests include pattern recognition, computer vision, and machine learning.



Xiangtao Zheng (Member, IEEE) received the M.Sc. and Ph.D. degrees in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China, in 2014 and 2017, respectively.

He is an Associate Professor with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China. His main research interests include computer vision and pattern recognition.



Xiaoqiang Lu (Senior Member, IEEE) is a Full Professor with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China.

His research interests include pattern recognition, machine learning, hyperspectral image analysis, cellular automata, and medical imaging.



Alexander A. Doudkin is a Professor with the Laboratory of System Identification, United Institute of Informatics Problems of the National Academy of Sciences of Belarus, Minsk, Belarus, and also with the Belarusian State University of Informatics and Radioelectronics, Minsk.

His research interests include digital signal and image processing, pattern recognition, architectures and models of computer vision systems.

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